System Performance Report

SafeEdge Edge-Based Fire Detection Analytics

Prepared for: Hackathon Judges & Stakeholders

March 01, 2026

SafeEdge

Edge-Based Fire Safety Intelligence System

DLW 2026 | Track 3: AI in Security | NTU Singapore

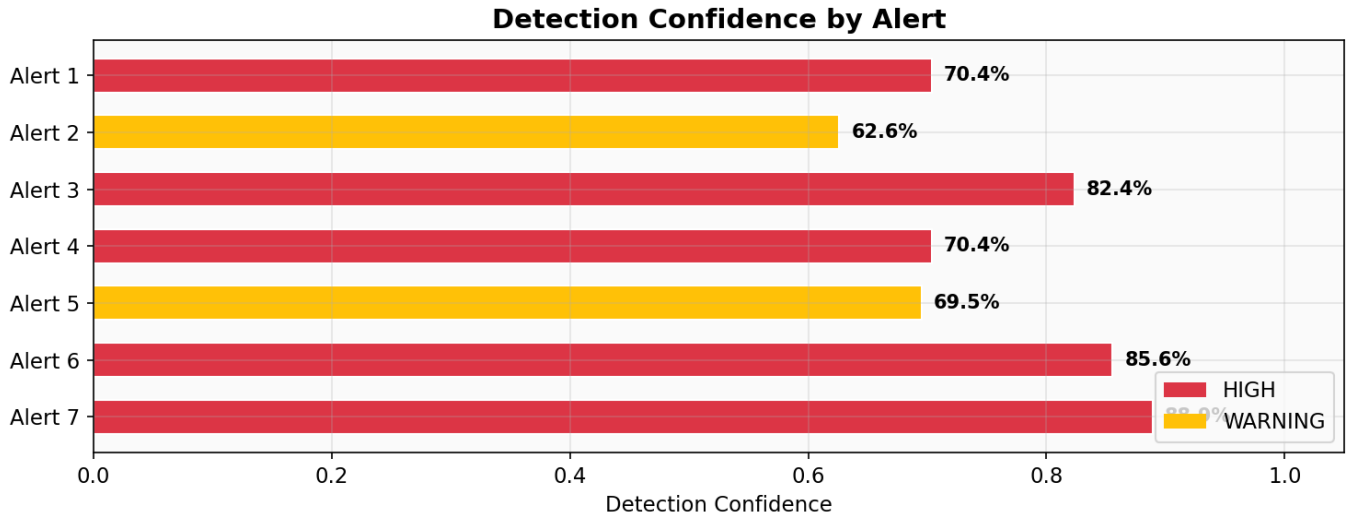
Executive Summary

In the recent analysis of fire safety alerts, a total of seven alerts were recorded, comprising three fire alerts and four smoke alerts. Of these, five were classified as HIGH risk, while two were designated as WARNING, indicating a significant level of concern regarding potential fire incidents. The average confidence level of the detections stood at 75.7%, with an average frame per second (FPS) rate of 8.2, suggesting reliable detection capabilities. All alerts originated from CAM_01, utilizing multi-frame confirmation with a positive detection rate of 5-8 out of 8 frames, underscoring the robustness of the surveillance system in identifying fire-related threats.

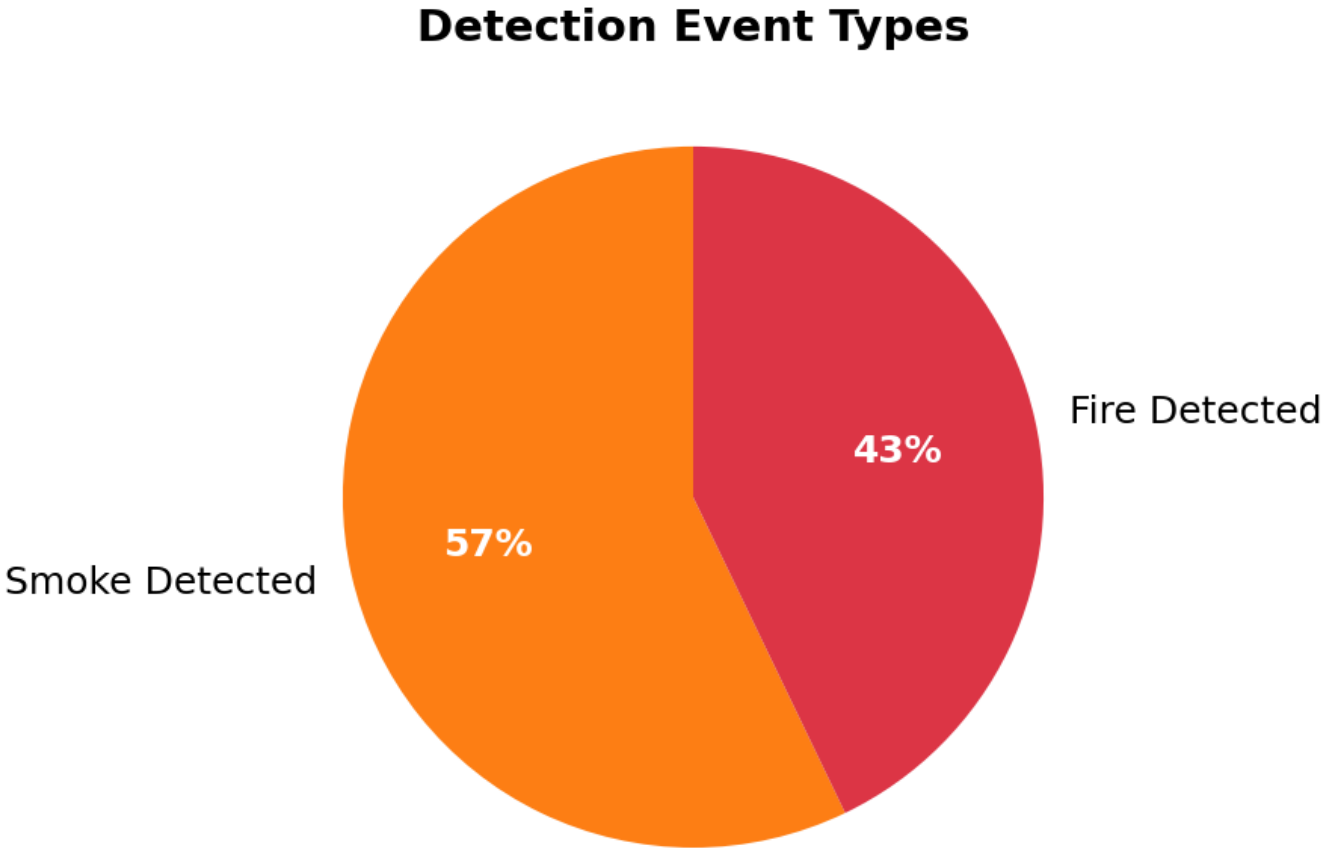
7	76%	8.2	5
Total Alerts	Avg Confidence	Avg FPS	HIGH Risk

Detection Performance

The detection performance analysis indicates a confidence range of 62.6% to 88.9% across various event types, including three fire alerts and four smoke alerts. All alerts were validated through a multi-frame sliding window approach, with 5 to 8 out of 8 frames confirming the presence of an incident. The risk assessment categorized five incidents as HIGH risk and two as WARNING, highlighting the critical need for prompt response measures in high-risk scenarios to mitigate potential hazards effectively.



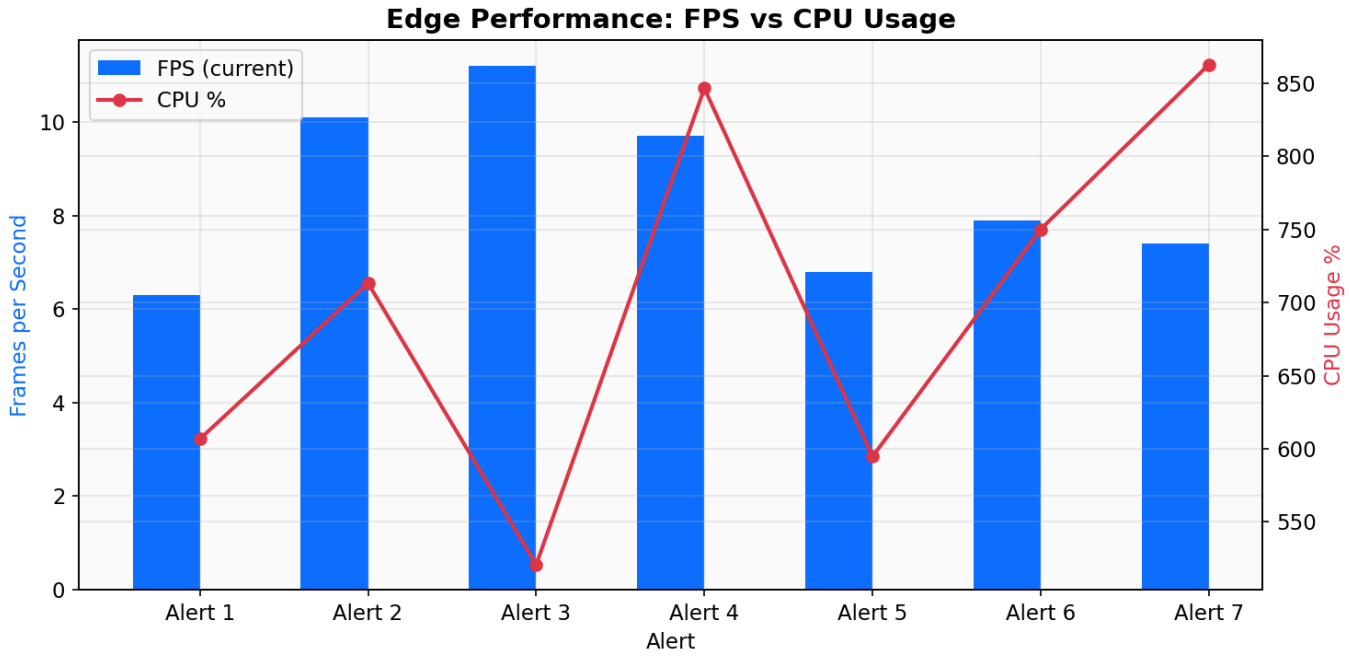
Detection confidence by alert, color-coded by risk level



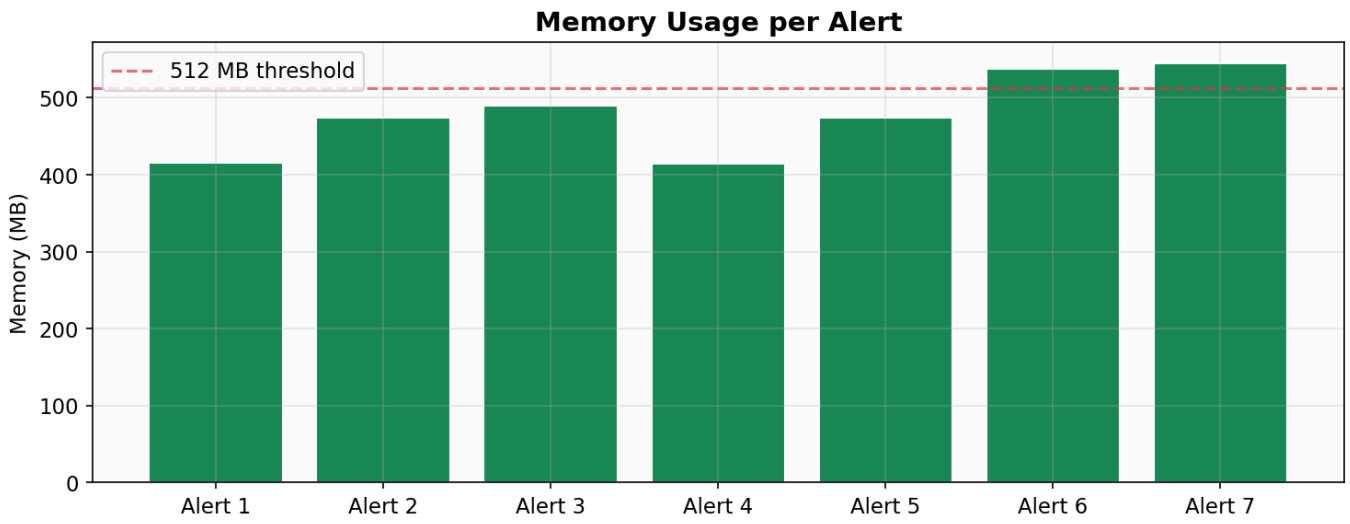
Distribution of detection event types

Edge Computing Metrics

The performance analysis of edge computing for the YOLOv8n model indicates a frames per second (FPS) range of 6.3 to 11.2, demonstrating a viable option for real-time applications on standard laptop hardware. The average CPU utilization reached 699%, while memory consumption averaged 479 MB, highlighting the efficiency of the model's implementation without the need for a GPU. With the model size at approximately 6 MB, these metrics suggest that edge computing can effectively support advanced inference tasks in resource-constrained environments.



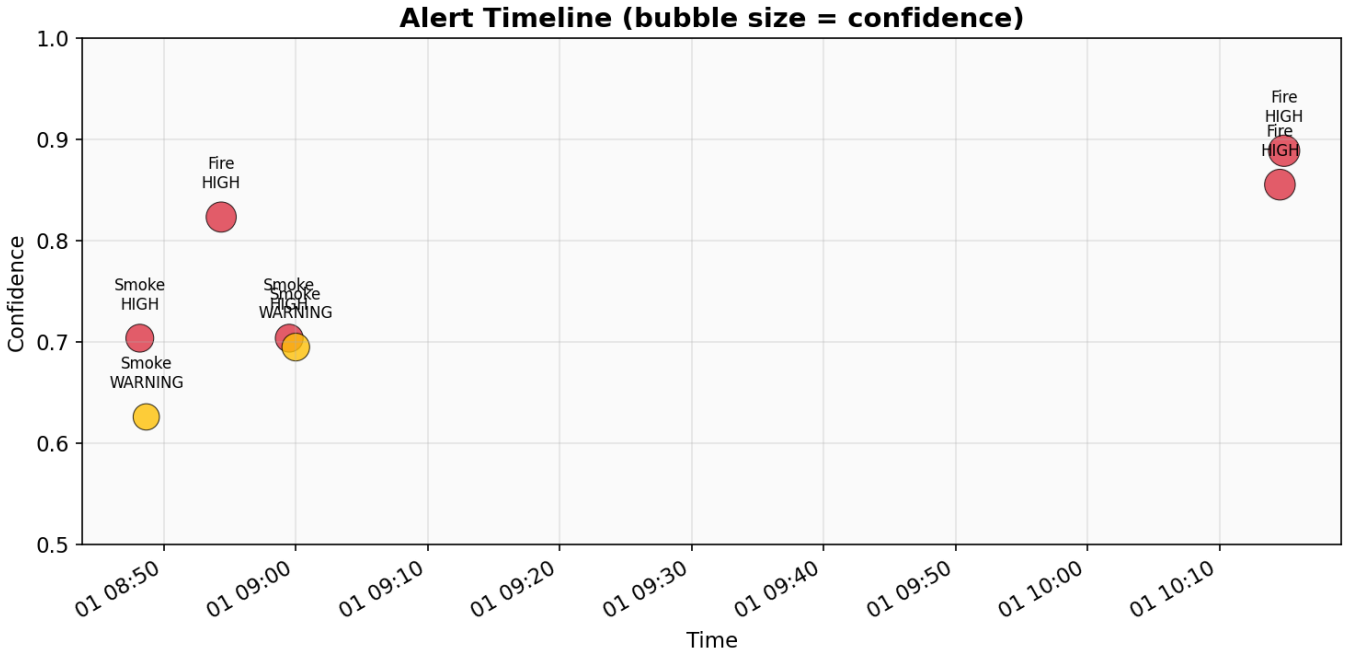
FPS throughput vs CPU utilization per alert



Memory footprint per detection alert

Alert Timeline & Escalation

The alert timeline analysis for the detection window from March 1, 2026, at 08:48:11 to 10:14:54 indicates a progressive escalation from smoke detection to fire confirmation across multiple buildings on the NTU campus, specifically the School of Chemical and Biomedical Engineering, Block 4A, and Northspine. The data suggests a critical window of approximately 86 minutes during which early intervention could have been pivotal in mitigating the fire's impact. This timeline underscores the importance of rapid response protocols and the need for continuous monitoring systems to enhance safety measures across campus facilities.



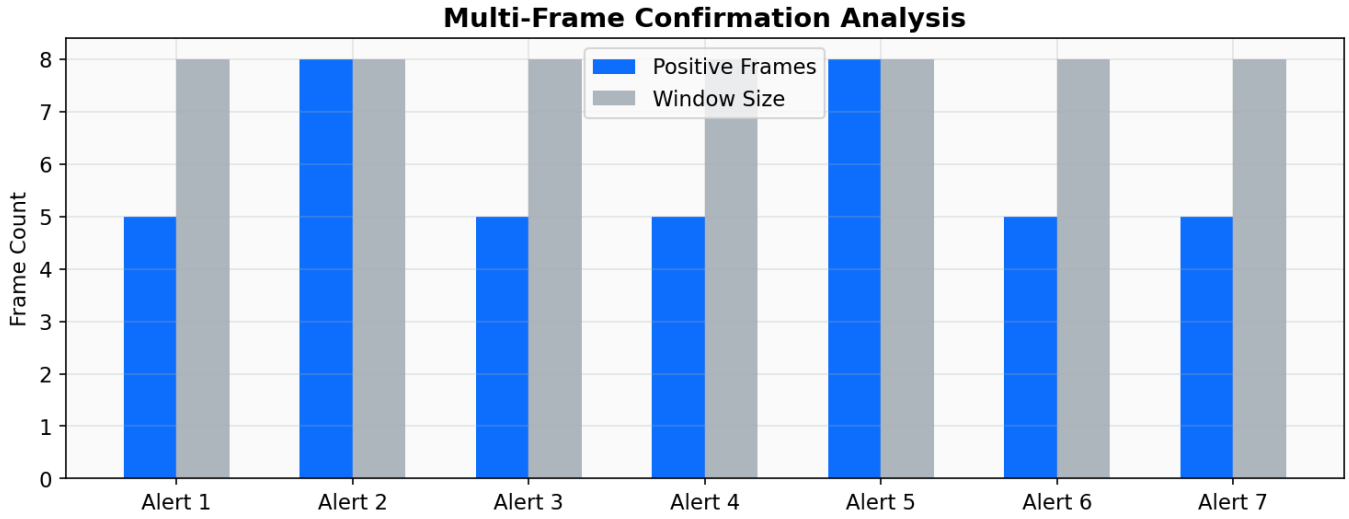
Alert timeline with confidence as bubble size

Alert Details

#	Event	Confidence	Risk	Location	Time
1	Smoke Detected	70.4%	HIGH	Block 4A F1	08:48:11
2	Smoke Detected	62.6%	WARNING	Block 4A F1	08:48:40
3	Fire Detected	82.4%	HIGH	Block 4A F1	08:54:21
4	Smoke Detected	70.4%	HIGH	Block 4A F1	08:59:30
5	Smoke Detected	69.5%	WARNING	Block 4A F1	09:00:00
6	Fire Detected	85.6%	HIGH	Northspine F1	10:14:35
7	Fire Detected	88.9%	HIGH	School of Chemical and Biomedical Er	10:14:54

Multi-Frame Confirmation

The implementation of a multi-frame confirmation system utilizing an 8-frame sliding window significantly enhances detection accuracy in fire safety monitoring. By requiring a confirmation threshold of 5 out of 8 frames to validate an event, the system effectively mitigates false positives caused by transient phenomena such as flickering lights and reflections. This robust approach ensures that only genuine fire incidents are flagged, thereby optimizing response efforts and resource allocation in critical situations. The reduction in false alarms not only conserves operational resources but also improves overall trust in the monitoring system.



Positive frames vs window size per alert